PROJECT DOCUMENTATION

Degree Program ECI
ASSIST HEIDI

Bluetooth Gaming Module for Natascha

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26.06.2026, Vienna

Abstract

In this assignment, it has aimed to show that a UART- controlled LED sequencer can be implemented on an STM32L432KC-Nucleo32 board using a FreeRTOS architecture. The system records a fixed input sequence (1, 1, 2, 2, 3, 3, 3) entered through a serial terminal, measures the elapsed time between each accepted input using the FreeRTOS osKernelGetTickCount(), and subsequently playbacks the recorded RGB LED timing pattern automatically during the second phase. Incoming UART characters are received via interrupt and placed into a message queue. The Control Task retrieves these characters from the queue, validates them against the expected sequence, and executes the corresponding action (LED switching and timing measurement) for each accepted character. Once the full sequence has been recorded, the measured timing values are passed to the Playback Task, which reproduces the recorded RGB LED pattern automatically during the playback phase.

The button is handled through a GPIO interrupt. When the button is pressed, the interrupt sets an event flag, which is checked by the Button ISR Task. Once the flag is set, this task forwards an abort command into the input queue, causing the Control Task to stop the current recording session and return to the start phase.

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1 Introduction

The objective of this project was to refine and optimize an existing assistive gaming station prototype developed for our co-designer, Natascha. The target games for this system are titles such as Blue Prince, an adventure, strategy, and puzzle-roguelike game played on PC, which is representative of a broader category of games requiring two simultaneous forms of input: camera perspective control, achieved through continuous mouse movement, and character navigation, achieved through discrete directional input via the WASD keys. Enabling fluid, responsive control over both of these input modalities was therefore a central requirement for the system to be genuinely usable for recreational gaming.

A previous iteration of the assistive gaming station successfully demonstrated that these keybinds could be mapped onto an accessible hardware layout suited to Natascha's needs. However, while functionally adequate, the final design of this earlier prototype introduced a number of physical and practical challenges that limited its everyday usability. The system as a whole was bulky and cumbersome, making it difficult to integrate cleanly into Natascha's existing wheelchair setup. Its operation relied on wiring spanning across the user's wheelchair setup, creating both a visual and physical obstruction and introducing a risk of accidental disconnection or interference with the chair's primary mobility functions. In addition, the prototype depended on external AAA battery compartments and loose elastic bands to secure its components in place, making the system cumbersome to mount, position, and transport independently, and undermining the goal of providing Natascha with a self-sufficient, easy-to-use setup. In light of these limitations, Natascha requested a streamlined, portable, and clean alternative that eliminated the mechanical clutter and physical bulk associated with the previous design, while preserving its core functionality.

To design an effective solution, the hardware structure had to align closely with the user's physiological realities, rather than being adapted from a generic input device after the fact.
Natascha lives with Cerebral Palsy. This condition results in spasticity, involuntary tremors, and severe muscle stiffness. Consequently, conventional PC gaming mice, keyboards, and standard gamepads are inaccessible to her, as they demand a level of fine motor precision and simultaneous multi-button coordination that is incompatible with these physical constraints.

For the system to be both usable and sustainable over extended play sessions, it had to leverage Natascha's existing physical strengths without causing rapid fatigue. Her core retained abilities include functional left-hand dexterity and controlled side-leg and knee movements, both of which informed the choice of input mechanism and its physical placement within the final design.

Together, these constraints framed the engineering problem not simply as one of building a wireless controller, but of designing a physically minimal, fatigue-conscious system that channels Natascha's specific retained motor functions into reliable, low-latency PC input. “**Figure 1.**” shows the logo of Team Probello, the team responsible for this project designed by it’s own members.

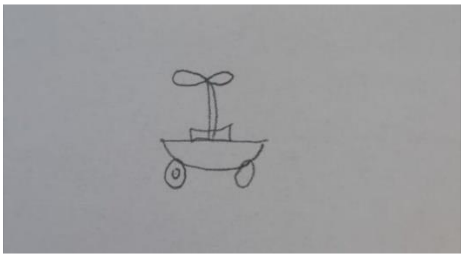

Figure 1. Flag for the Team Probello

“**Figure 2.**” presents the block diagram of the overall system, illustrating how the wheelchair-mounted input hardware, the wireless mode-switching module, and the host PC interact to deliver both mouse and keyboard-style control from a single, compact, and self-contained setup.

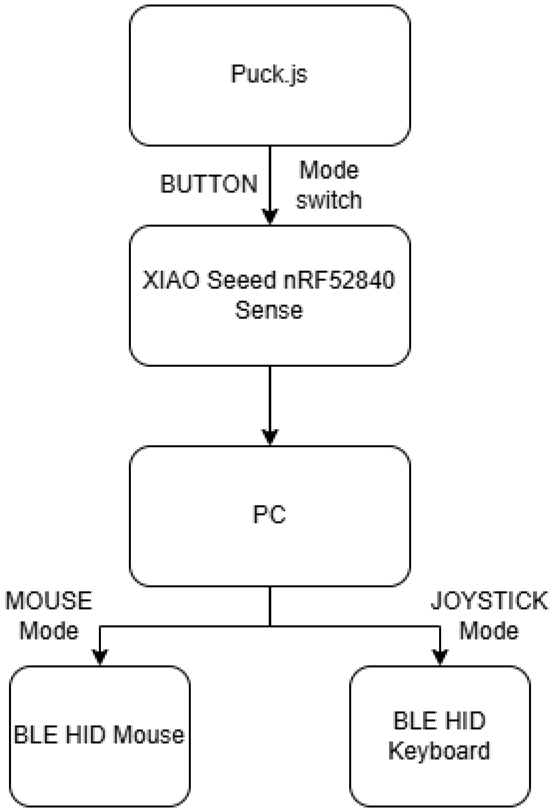

Figure 2. Block diagram for the whole system

2 Methods

To address these constraints, Team Probello, working in close collaboration with the co-designer, brainstormed potential approaches and ultimately engineered a wireless, dual-node assistive setup that eliminates all physical cable connections between the user's hand inputs, leg inputs, and the gaming PC. By treating the hand input and the leg input as two isolated, self-powered wireless units rather than a single tethered assembly, the resulting controller drastically reduces setup time and removes the physical bulk that had previously accumulated on Natascha's wheelchair.

The architecture consists of two main components. The first is the Acceleration-Detecting Joystick, referred to as the Hand Node, which is housed in a custom shell tailored specifically to Natascha's left-hand dexterity. At the core of this node is a Seeed Studio XIAO nRF52840 Sense microcontroller, shown in “**Figure 3.**”, which integrates a Bluetooth Low Energy radio with an onboard accelerometer and Inertial Measurement Unit (IMU) sensor on a single, compact board.

Figure 3. Photo of the used microcontroller XIAO Seeed Studio nRF52840 Sense

This MCU continuously detects the physical acceleration of the joystick as it is moved and emulates these movements directly as PC Human Interface Device (HID) outputs, namely WASD key presses for character navigation and mouse movement for camera control.

The second component is the Puck.js Switch, referred to as the Knee Node.

“**Figure 4**” shows the Puck.js module used for this purpose, coin-sized Bluetooth Low Energy development board fitted into a custom mechanical fixture originally developed by the previous HEIDI group working with Natascha, the Trentino Team.

Figure 4. Photo of the used button, as a mode switcher between MOUSE and KEYBOARD

Positioned directly next to Natascha's wheelchair frame, this fixture is built to respond to her left knee and leg side movements, allowing her to trigger a mode switch using a part of her body distinct from the hand controlling the joystick itself.
"**TABLE 1.**" demonstrates the desired inputs and expected outputs, as associated action for the project.

TABLE 1. Table of controls and associated actions

Physical Input Device	User Action	System Output	Associated Action
Puck.js Knee Switch	Single Press (Left Knee)	Mode Swap Toggle	Cycles between Keyboard Mode and Mouse Mode
Smart Joystick (Keyboard Mode)	Move Left / Right / Forward / Backward	Arrow Keys (W-A-S-D)	Controls character navigation
Smart Joystick (Mouse Mode)	Move Left / Right / Forward / Backward	Mouse Cursor Movement	Controls camera perspective and aiming

The following table provides a comprehensive overview of the Bill of Materials (BOM) and sourcing details required for the assembly of the system. "**TABLE 2.**" illustrates the complete breakdown of the technical components, categorized into primary microcontrollers, power management, wiring interfaces, and essential production tools. Notably, the hardware architecture relies on the Seeed Studio XIAO nRF52840 and a Puck.js Bluetooth smart beacon to handle the core processing and connectivity.

TABLE 2. BOM list for the components on the project

Item	Category	Part Description / Notes	Qty	Manufacturer / Model	Remarks
1	Main Hardware	Microcontroller (MCU) Features integrated Bluetooth LE, internal LiPo charging circuit, and Type-C port.	1	Seeed Studio XIAO nRF52840	
2	Main Hardware	Knee Node ProcessorBluetooth Smart Beacon.	1	Puck.js	
3	Power & Wiring	Power Source 3.7V / 170mAh rechargeable LiPo battery.	1	HWE601525	
4	Power & Wiring	Charging Extension USB-C Male-to-Female Extension Cable (creates a rugged exterior charging socket).	1	-	
5	Power & Wiring	Toggle Interface SPST Sub-miniature slide switch.	1	-	
6	Power & Wiring	Interconnects Thin-gauge internal hookup wire (fitted with a quick-disconnect plug for clean teardowns).	1	-	
7	Tools & Consumables	Tools Required Soldering iron, wire strippers, heat-shrink tubing, glue.	-	-	Production Tool Only
8	Reused Legacy Assets	Joystick Cap Inherited for core mechanical alignment.	1	Reused Asset	Group Trentino
9	Reused Legacy Assets	Silicone Rubber Cover Provides waterproofing and sweat resistance for the joystick interface.	1	Reused Asset	Group Trentino
10	Reused Legacy Assets	Wheelchair Frame Mount Side holder retained to clamp the knee node securely to the frame.	1	Reused Asset	Group Trentino

“**Figure 5.**” Illustrates the housing for the knee button. Note that Puck.js button will be placed on this, in order to keep it stable and let Natascha to switch between MOUSE and KEYBOARD modes instantly.

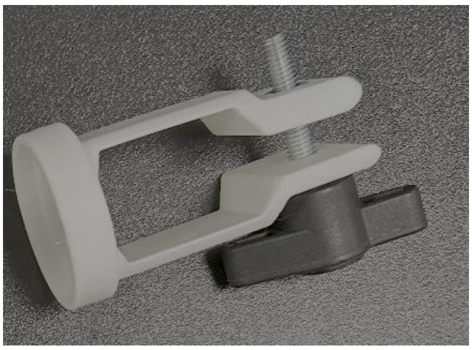

Figure 5. Housing for the knee button. (Puck.js)

“**Figure 6.**” shows the whole materials, including MCU XIAO Seeed Studio nRF52840 Sense, Joystick cap, Joystick body, USB-C extension cable, Battery and Switch.

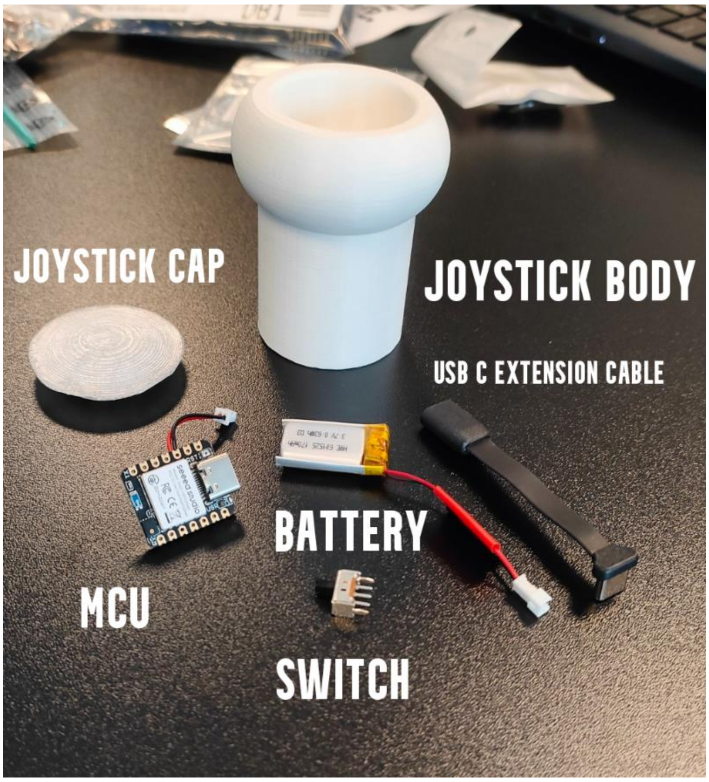

Figure 6.Joystick body along with its components (without the silicone cap)

2.1 3D Design

YOU SHOULD WRITE 3D PROTOCOL HERE.

2.2 Code Design (MAYBE WE CAN REMOVE THIS COMPLETELY FOR OUR DOCUMENTATION, BUT I THINK WE SHOULD MENTION THAT WE WRITE OUR CODES THROUGH ESPRUINO WEB IDE, AND ARDUINO IDE.

3 Results – Testing

After uploading the code, results have been observed using PuTTY serial terminal.
"**Error! Reference source not found.**." shows the screenshot for the terminal output, when there is no Button ISR or unexpected inputs.

4 Discussion